Reasoning-trace analysis — figures and findings

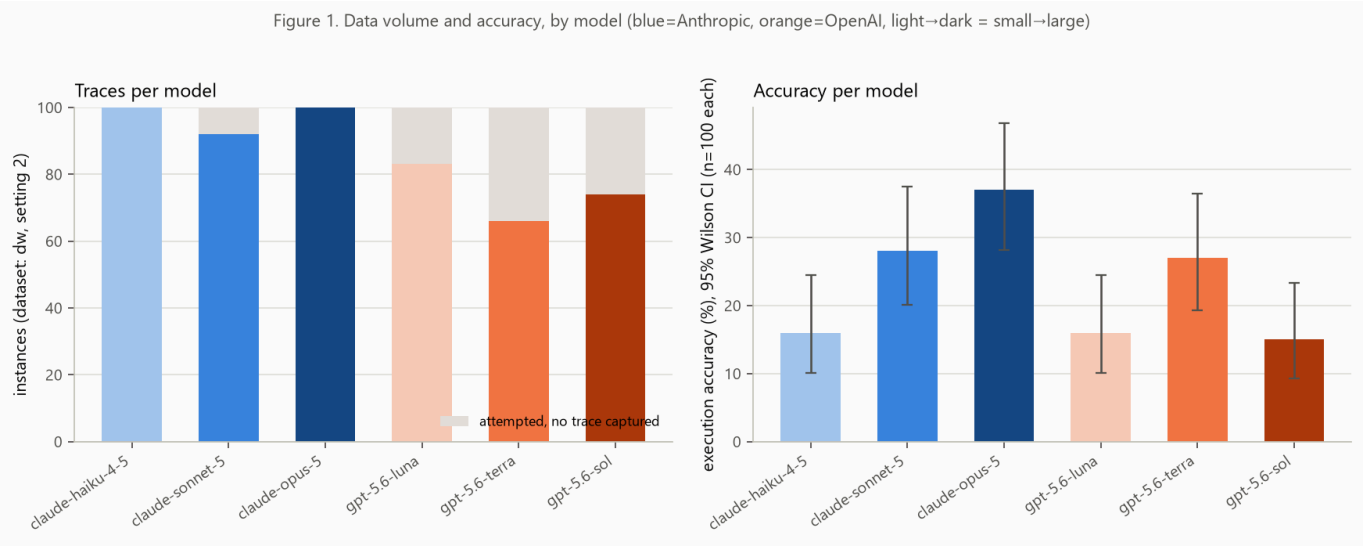
Updated 2026-09-26: pipeline re-run after an apostrophe-normalization fix; Figures 1, 7, 11, 12 redrawn; move-classifier agreement corrected (κ=0.69, see Figure 9). Previous version archived in `outputs/archive_20260925/`.

Companion to `report.md` (the prose-first version). This version walks through the 14 figures in order, each with its image and explanation inline. Same underlying numbers throughout — see `report.md` for the more compact narrative and `outputs/tables/` for the raw data behind every figure.

Scope: dataset `dw`, setting 2, 6 models (`claude-haiku-4-5`, `claude-sonnet-5`, `claude-opus-5`, `gpt-5.6-luna`, `gpt-5.6-terra`, `gpt-5.6-sol`) + 5 human reference rationales. 600 attempted instances, 515 with a captured reasoning trace, 47 with a trace from all 6 models. No LLM prompting or LLM-based labeling anywhere — all features are lexical, embedding-based, or a small trained classifier.

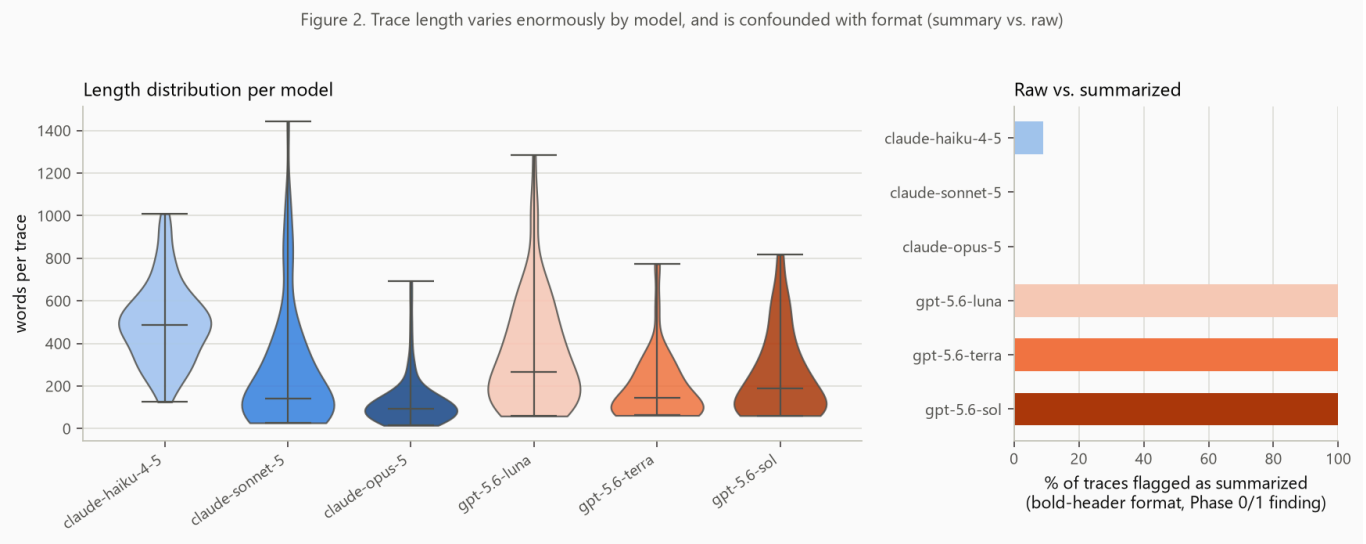
Read this before anything else: brand and trace *format* are almost totally confounded in this dataset — 100% of OpenAI traces are a bold-header summary (the signature of the Responses API's `reasoning.summary`) vs. 0–9% of Anthropic's, which are the model's actual raw extended-thinking text. Every "brand" finding below is at least partly a "raw vs. summarized" finding, and the two can't be statistically separated here (see Figure 5's caveat and the Limitations section).

Figure 1 — How much data per model, and how accurate is each?



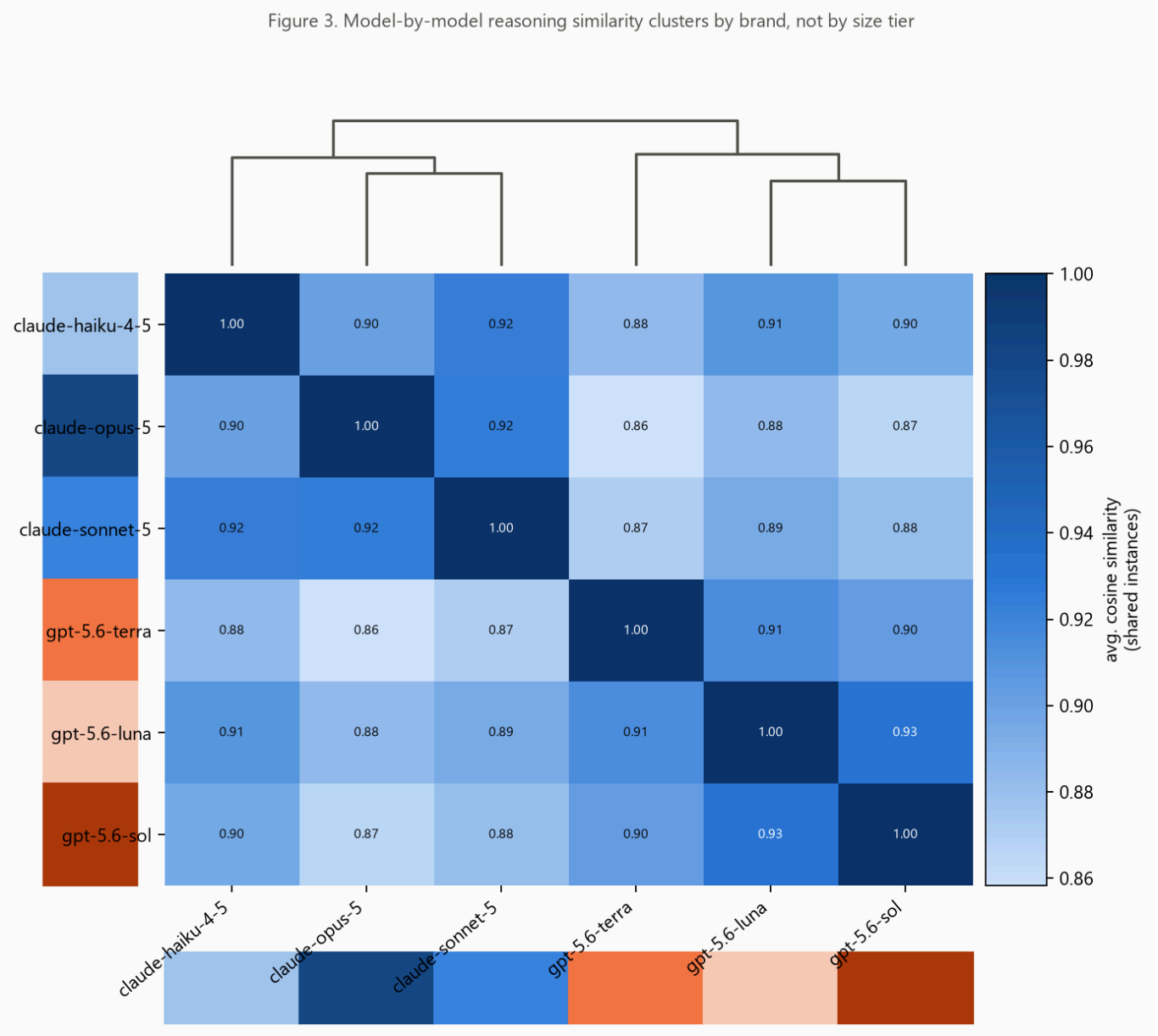
Accuracy ranges from 15% (`gpt-5.6-sol`) to 37% (`claude-opus-5`); the whiskers are 95% Wilson intervals (n=100 each, roughly ±8–10 points), so only large gaps mean anything — opus vs. haiku/luna/sol is clear; opus vs. sonnet/terra is not. Trace coverage (the gray segment = attempted but no reasoning trace captured) is 100% for `claude-haiku-4-5`/`claude-opus-5`, down to 66% for `gpt-5.6-terra`. That missingness is *not* random with respect to correctness for `gpt-5.6-luna` and `gpt-5.6-terra` specifically — the API returns a captured trace for only 56% of their correct answers vs. 88%/70% of their incorrect ones, even though `reasoning_tokens` confirms real internal reasoning happened either way. Correctness-conditioned results for those two models should be read as lower-confidence than the rest.

Figure 2 — How long are traces, and how does that relate to format?



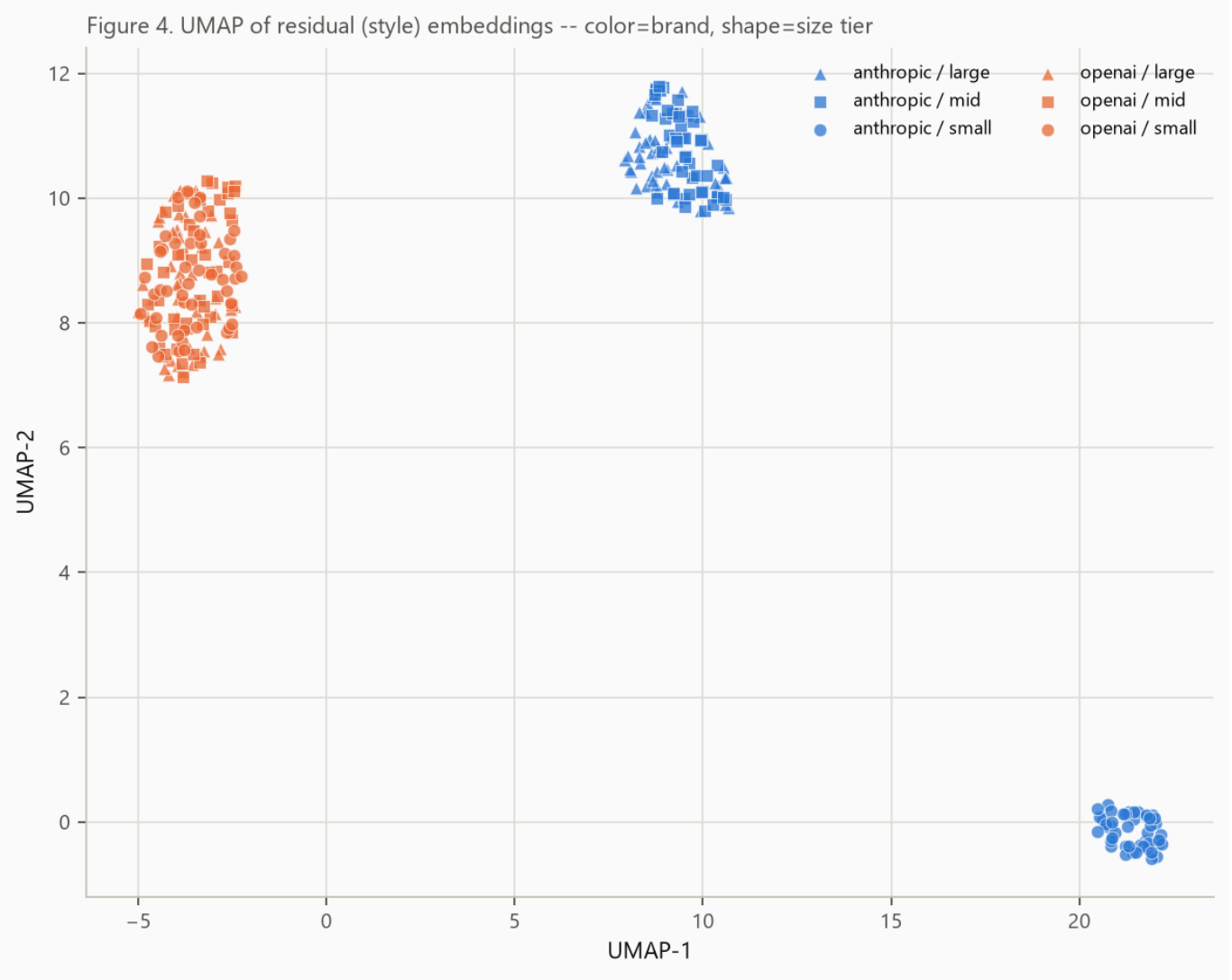
Trace length varies by an order of magnitude: `claude-opus-5` traces have a median of ~93 words, `claude-haiku-4-5` a median of ~488 (words counted after stripping markdown headers/code blocks). The right panel shows exactly the brand/format confound described above — essentially every OpenAI trace is a summary, essentially no Anthropic trace is (haiku is the one partial exception, at ~9%). Length and format are themselves entangled: the two shortest, most terse models (`opus`, `sonnet`) are also the two with almost no visible "process" language, which matters for how to read Figures 8–9.

Figure 3 — Do models cluster by brand or by size?



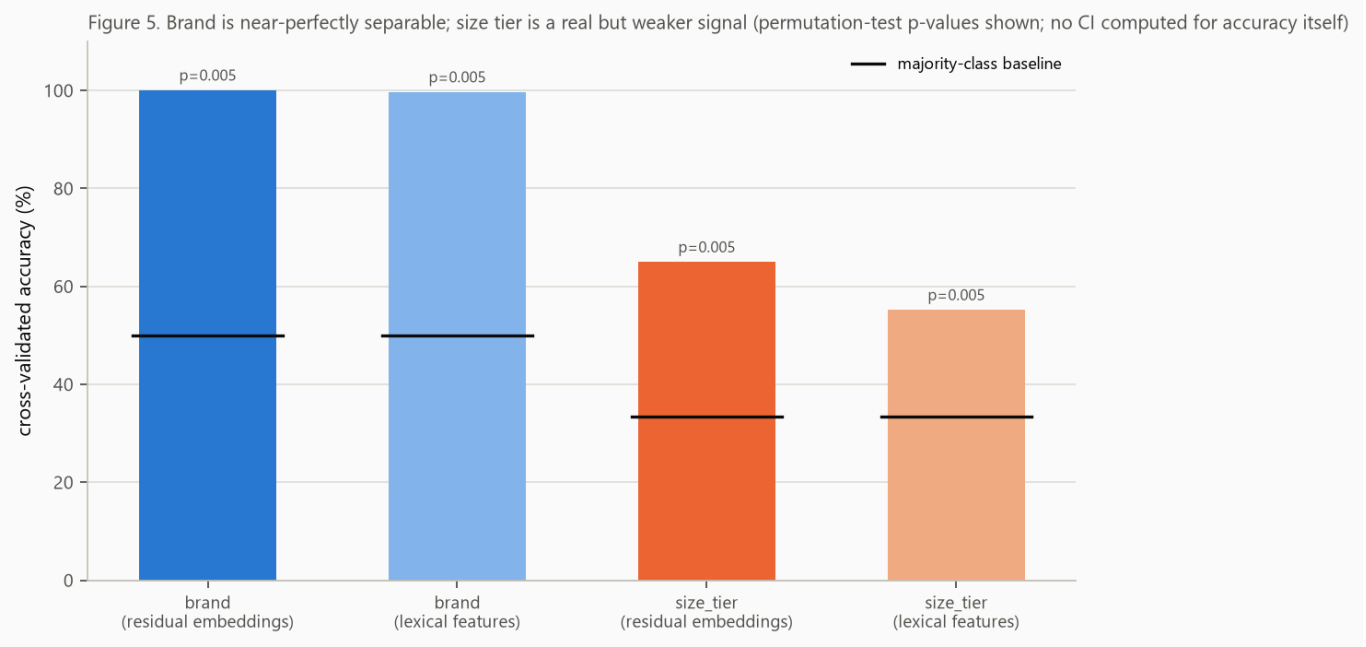
Hierarchical clustering on shared-instance trace-embedding similarity splits cleanly into two branches — and they're exactly the Anthropic/OpenAI split, before the dendrogram ever sees a brand label. Within a brand, pairwise similarity is 0.90–0.93; across brands it is 0.86–0.91 — the ranges touch (`haiku`–`luna` is 0.909), so the split is a clear tendency, not a hard gap. Clustering follows brand, not size tier — e.g. `claude-opus-5` (large) is more similar to `claude-haiku-4-5` (small, same brand; 0.90) than to any OpenAI model (0.86–0.88).

Figure 4 — Same question, in 2D, with instance content removed



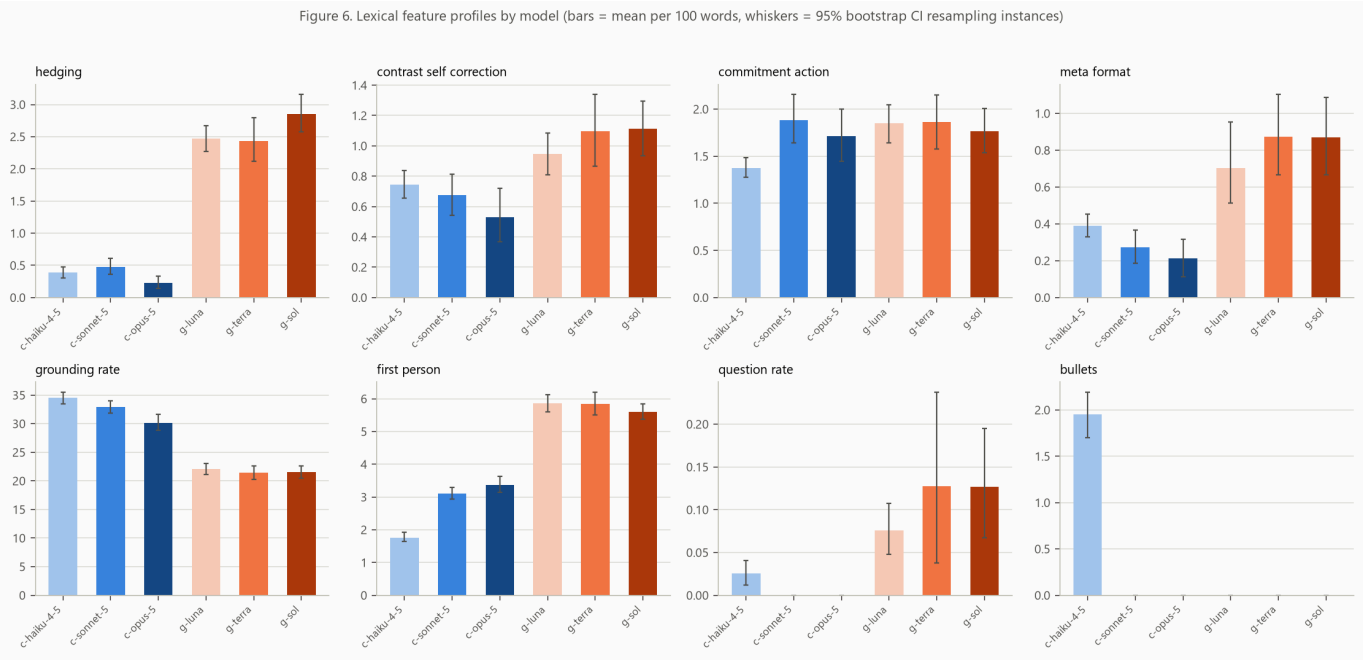
This is on *residual* embeddings — for each instance, the mean embedding across all 6 models is subtracted out, isolating style from content. Brand (color) still separates cleanly even after removing what's specific to each question. There are three clusters, not two: `claude-haiku-4-5` (all circles, bottom right) sits by itself, `claude-opus-5` + `claude-sonnet-5` (triangles/squares, top) form a second cluster, and all three OpenAI models are one intermixed blob — the plot cannot tell `luna`, `terra`, and `sol` apart. (Don't read cluster *size* or the distance between clusters as meaningful; and note that overlap in 2D hides real differences: opus vs. sonnet look mixed here but are ~91% separable in the full 1024-dimensional space. See `figure_guide.md`.)

Figure 5 — How separable is brand vs. size, from embeddings vs. lexical features?



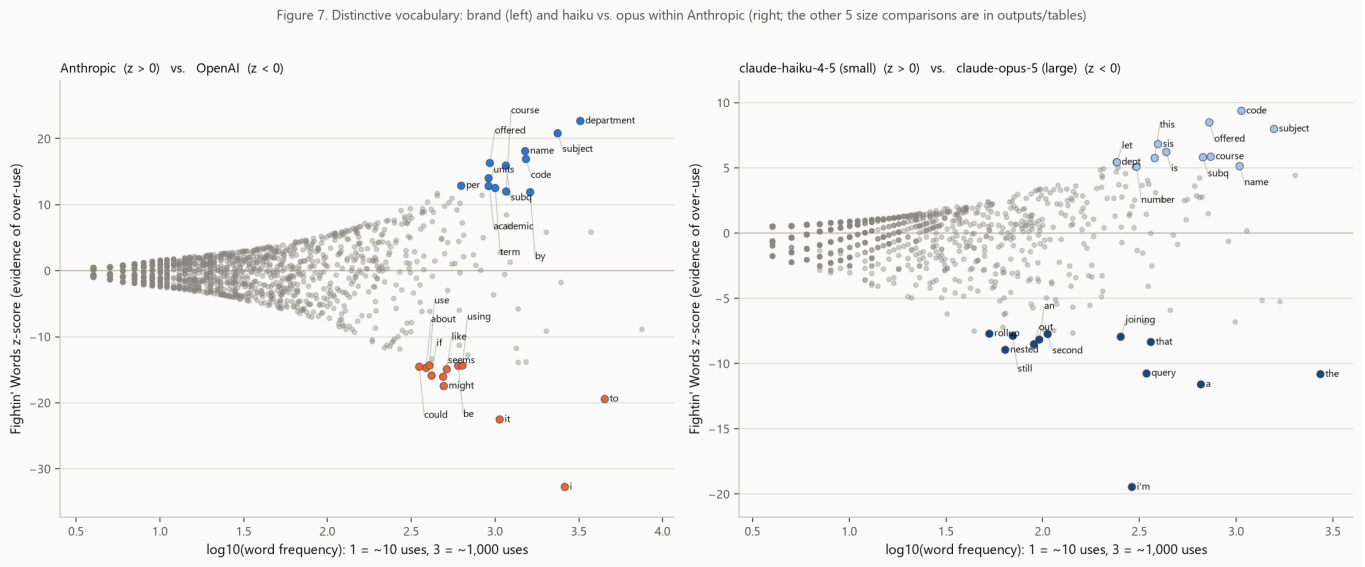
A logistic-regression probe (GroupKFold by instance, permutation-tested) predicts **brand at 100% (embeddings) / 99.6% (lexical features)** accuracy against a 50% majority baseline — both p=0.005. The single strongest lexical driver of that result is `is_summary` itself, confirming this is substantially the format confound, not a subtle stylistic signature. **Size tier is separable too, but far more weakly**: 65% (embeddings) / 55% (lexical) vs. a 33% baseline, still p=0.005 — a real signal, just nowhere near as strong as brand. No CI is shown on the bars themselves (only the permutation p-value was computed for accuracy; no bootstrap CI on accuracy was run).

Figure 6 — Per-feature style profile per model, with uncertainty



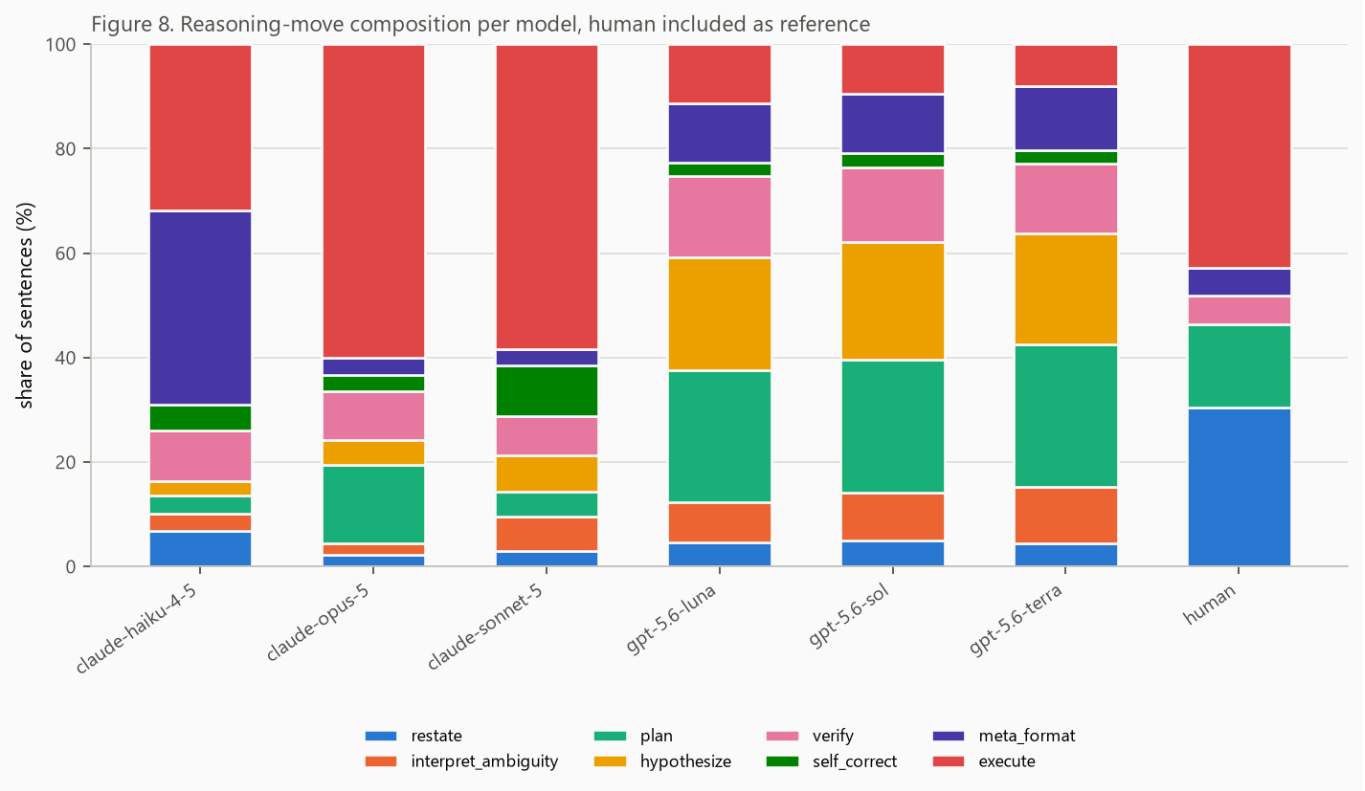
Eight lexical rate features (per 100 words), one per panel, with 95% bootstrap CIs (resampling instances within each model). `grounding_rate` — the share of tokens matching actual schema/prompt vocabulary — is visibly higher for Anthropic models, consistent with raw thinking walking through the real schema rather than narrating about it. `first_person` and `hedging` trend higher for OpenAI's narrated summaries.

Figure 7 — Which specific words distinguish brand / size?



Log-odds with an informative Dirichlet prior (Monroe et al.). Left: brand — words like `department`, `subject`, `name`, `code`, `offered` skew Anthropic (schema-grounded vocabulary), while `might`, `seems`, `if`, `i` skew OpenAI (hedging + first-person narration). Right: a representative size comparison (`claude-haiku-4-5` vs. `claude-opus-5`, within-brand small vs. large) — the other 5 possible size comparisons (openai small/mid/large, anthropic mid) are in `outputs/tables/fightin_words_size_*.csv` but not separately plotted here.

Figure 8 — What mix of reasoning moves does each model (and human) use?

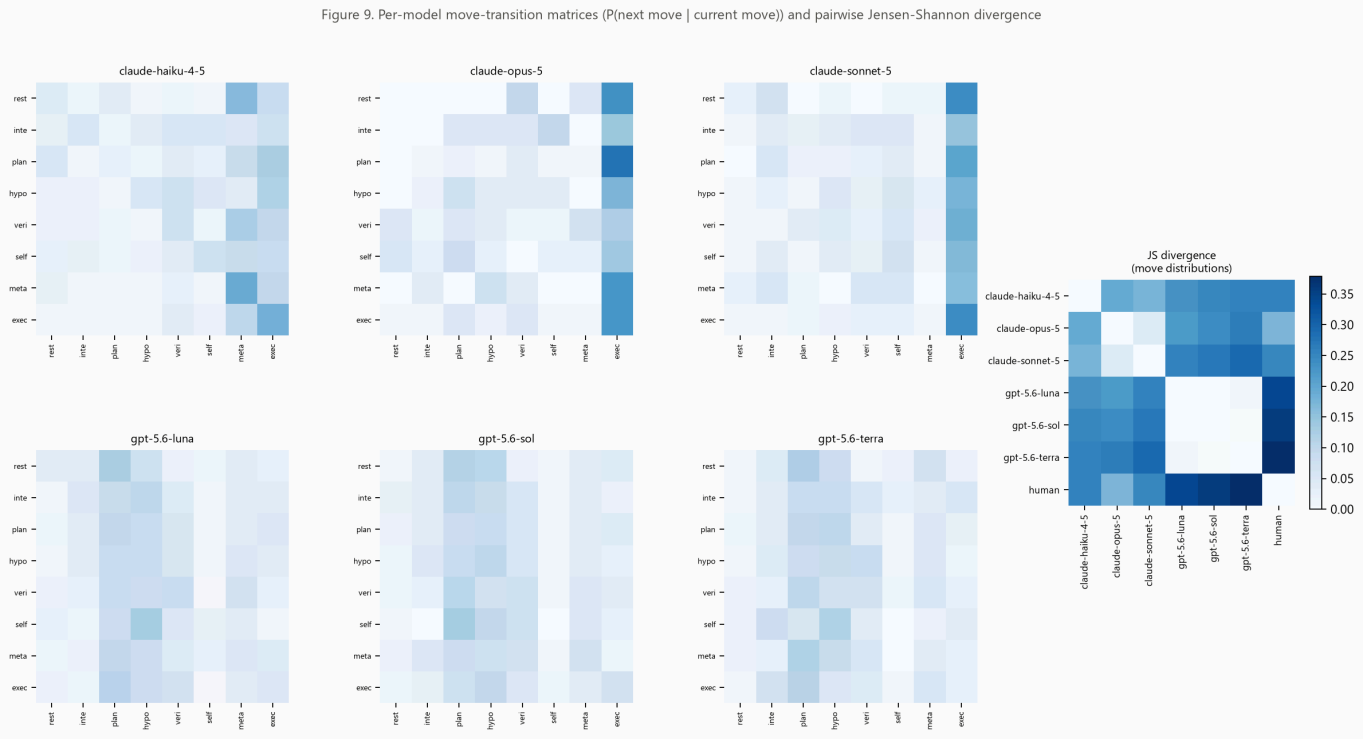


Eight bootstrapped move categories (restate / interpret_ambiguity / plan / hypothesize / verify / self_correct / meta_format / execute), human included as the rightmost reference bar. `claude-opus-5` and `claude-`

`sonnet-5` are extremely `execute`-dominated (58–60%) with little else — almost certainly a **length artifact** (they're also the two shortest-trace models from Figure 2; a 5-sentence trace mostly *is* "writing the query"), not necessarily a distinct reasoning style. `claude-haiku-4-5` (much longer traces) shows a far more diverse mix. The `gpt-5.6-*` models spread evenly across plan/hypothesize/verify/meta_format/execute.

The single most striking number in this whole analysis is here: **humans restate the question in ~30% of their sentences (0.304); no model exceeds 6.5%.** Every model, regardless of brand or format, frames the problem far less than a human does when explaining their own reasoning.

Figure 9 — What follows what, and how far apart are models' move usage?

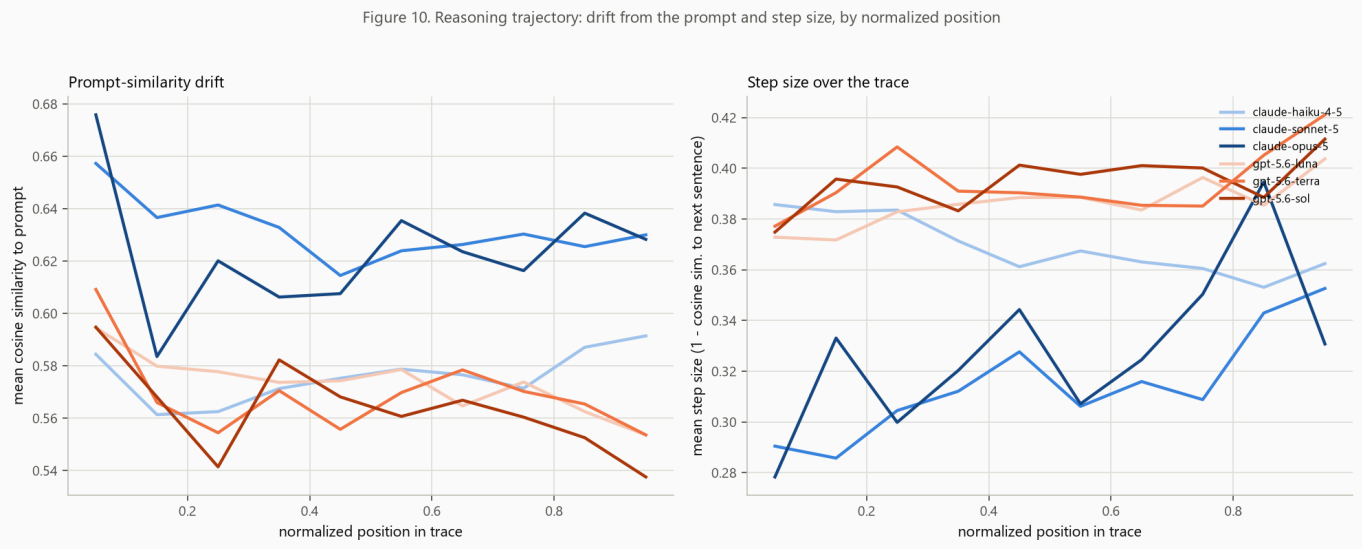


Left 6 panels: per-model P(next move | current move). The terse Anthropic models show heavy "→execute" columns; the OpenAI models show more spread-out transitions centered on "→plan". Right: pairwise Jensen-Shannon divergence between every model's (and human's) move distribution. The three `gpt-5.6-*` models are nearly identical to each other (JS 0.001–0.005); Anthropic-vs-OpenAI pairs sit around 0.22–0.29; and **`human` is the single most distinct row in the entire matrix** (JS 0.17–0.38 against every model) — human explanations are compositionally unlike any of the 6 models, not just unlike one brand.

Data-quality note (revised 2026-09-26): the underlying move classifier initially had a real bug — an unweighted classifier collapsed `restate` and `self_correct` (the two rarest seed categories) to zero predictions. Fixed with `class_weight="balanced"`. **My earlier claim that agreement with your hand labels "held at κ=0.94" was wrong**: that check compared your labels to the predictions stored in the validation file at export time (the pre-fix classifier), not to the classifier behind these figures. Scored properly against the current classifier: **Cohen's κ = 0.69 (73.8% raw agreement, n=195 of 200)**. Per move: the four large moves `interpret_ambiguity`, `plan`, `hypothesize`, `meta_format` have F1 0.74–0.82; `execute` has precision 1.00 but recall 0.63; `verify` recall 0.59; and the classifier now *over-predicts* the rare moves (`self_correct` for 12 of your 200 sentences, which you labeled once; `restate` for 5, labeled zero times). Excluding both rare moves, κ = 0.76. Cautions: your sample was drawn from the old classifier's predictions and

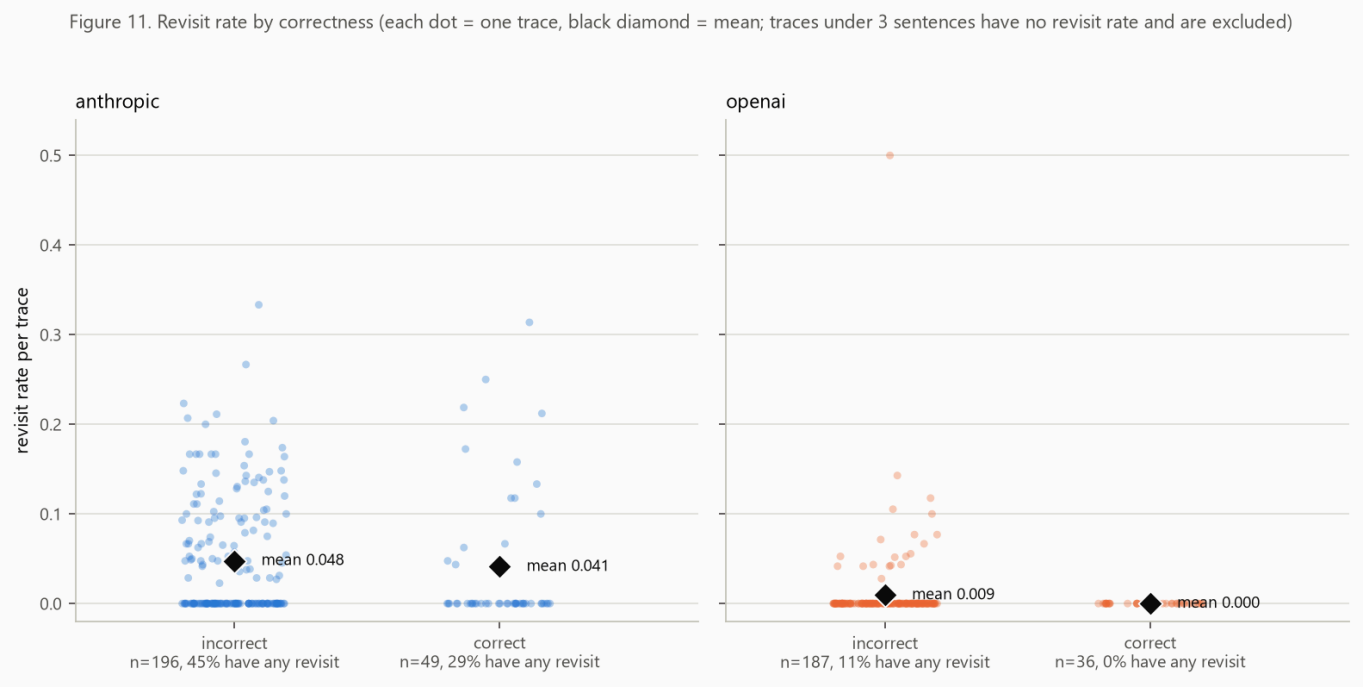
shown beside your label column (possible anchoring), and it is not a fair test of the two rare moves. So: model-level shares of the four large moves are reasonably trustworthy; `restate`/`self_correct` shares (and the `self_correct` result in Fig 12) are unvalidated and probably inflated. Human sentences were never validated — my own spot check of the 17 human sentences labeled `restate`: ~13 read as genuine restatements, ~4 look like `plan` (~23% rather than 30.4%; still ~3.5× the highest model, 6.7%).

Figure 10 — Does reasoning drift from the prompt, and how fast?



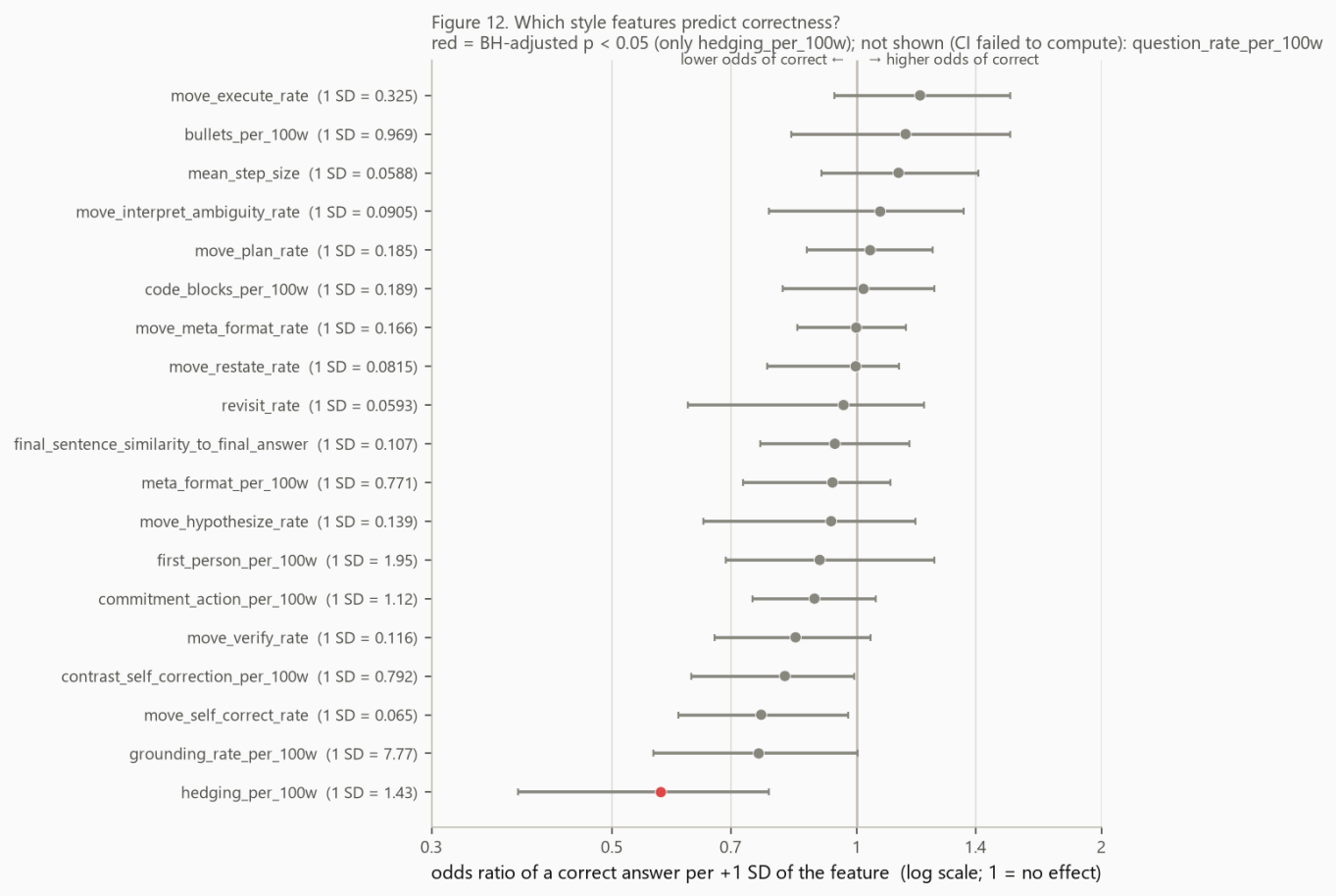
Left: cosine similarity between each sentence and the original prompt, over normalized position in the trace (0=first sentence, 1=last). Right: step size (1 − cosine similarity to the next sentence) over the same normalized position. This does **not** split by brand. The two terse Anthropic models (`opus`, `sonnet`) have the highest prompt similarity (~0.60–0.68) and the smallest step sizes early on (~0.28–0.30, rising toward the end); `haiku` and all three OpenAI models sit lower on prompt similarity (~0.54–0.61) and take larger, steadier steps (~0.35–0.42). Treat this figure as weak evidence: the lines are noisy with no CIs drawn, `opus` traces have only ~4 sentences (so each bin is 1–2 sentences), and "similarity to the prompt" is measured against only the first 512 tokens of a ~13,000-token prompt (the embedding model's input limit) — mostly the generic instruction boilerplate, not the question or schema.

Figure 11 — Does revisiting earlier reasoning associate with correctness?



Revisit rate = share of sentences whose closest match is a *non-adjacent* earlier sentence (similarity > 0.884, the 90th percentile of that within-trace distribution — tuned, not a fixed constant). Split by correct/incorrect within each brand; each dot is one trace, the black diamond the mean. Within Anthropic the *means* are similar (0.048 incorrect vs. 0.041 correct), but the **share of traces with any revisit at all is 45% for incorrect (89/196) vs. 29% for correct (14/49)**. A naive two-proportion test gives p≈0.03, but it ignores that traces of one problem are correlated and is not part of the corrected 20-feature battery (revisit rate as a continuous feature: p_BH=0.96) — a hint, not a finding. For OpenAI, all 36 correct traces have a revisit rate of exactly 0 vs. 20 non-zero cases among 187 incorrect ones. Traces under 3 sentences (31 `opus`, 16 `sonnet`) have no defined revisit rate and are excluded.

Figure 12 — Which style features actually predict correctness?



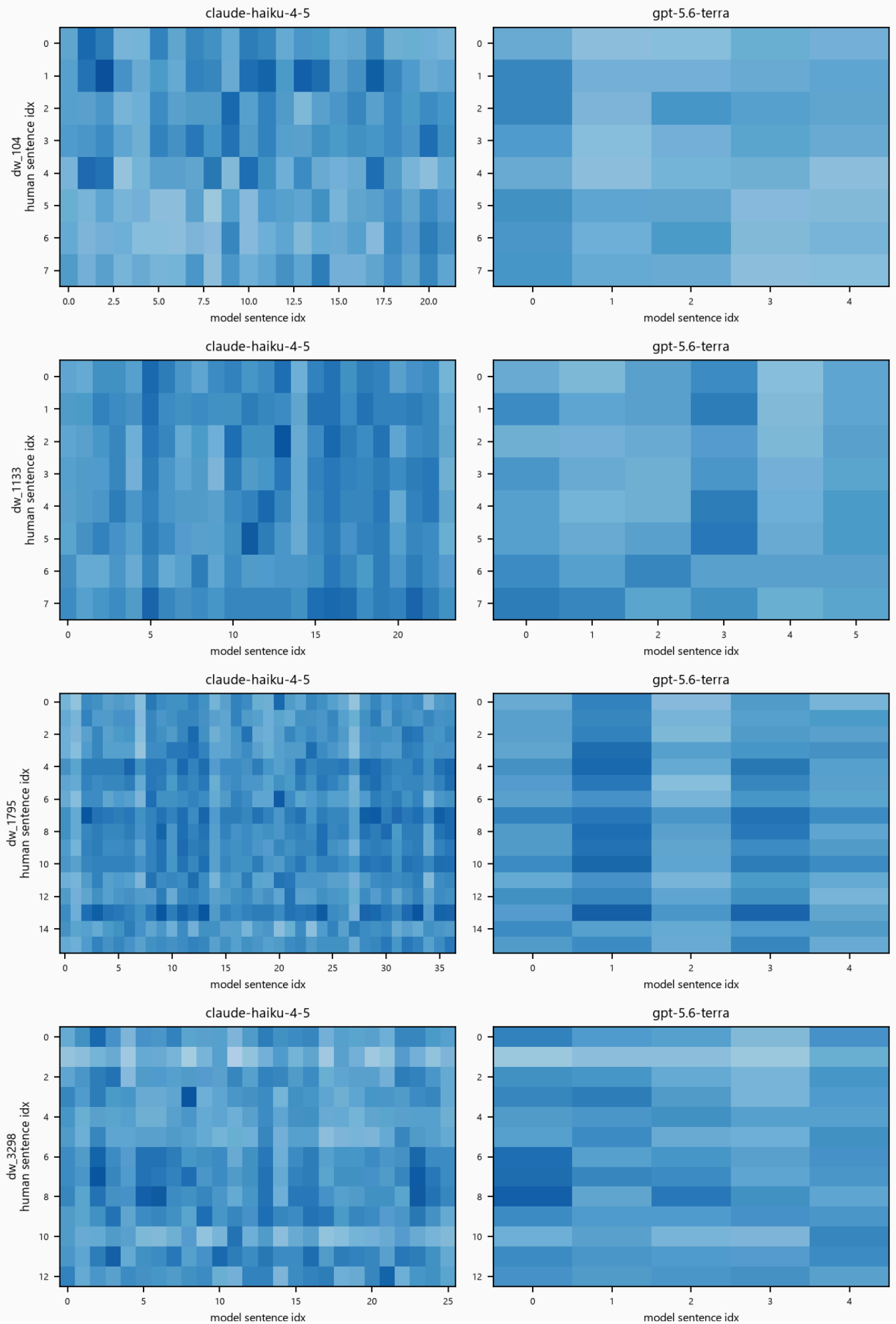
(Redrawn 2026-09-26: the x-axis is now the odds ratio per +1 standard deviation of each feature, on a log scale, so rows are comparable; the previous version plotted raw coefficients whose scales differed by feature.)

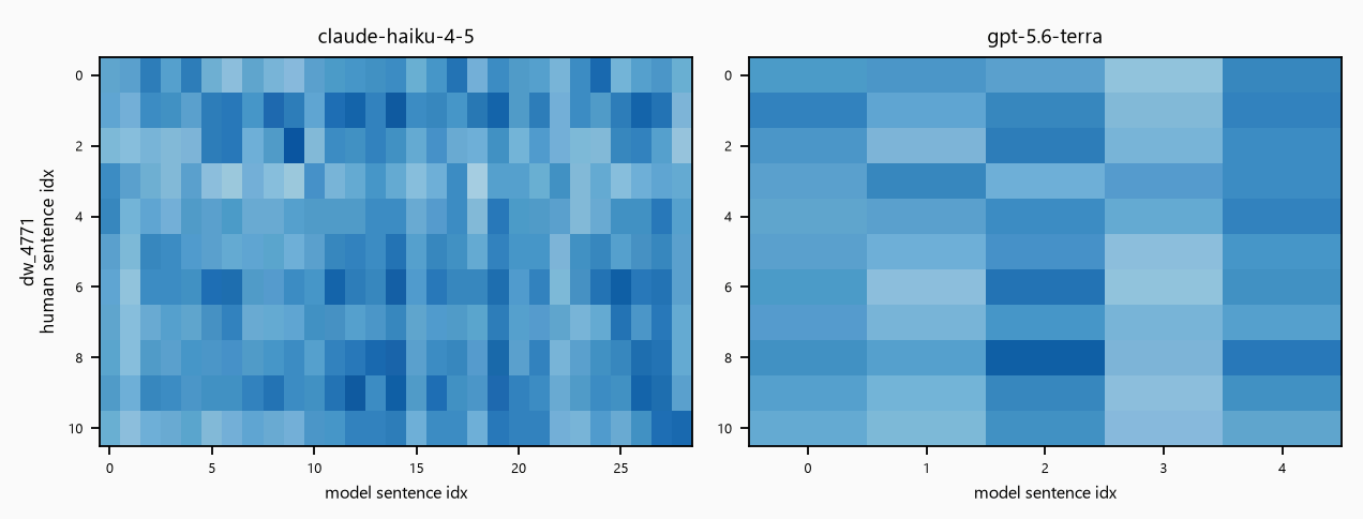
20 style features tested via `correct ~ feature + brand + size_tier` (GEE, grouped by instance), 95% CI from 200 bootstrap resamples of *instances*, and Benjamini-Hochberg correction across the family. **Only `hedging_per_100w` survives correction** (red dot): coefficient −0.39, 95% CI [−0.67, −0.17], p_BH=0.031, n=515 — more hedging predicts *lower* correctness. This only reached significance at full data scale (it didn't clear correction on a 50-per-model dev sample). `move_self_correct_rate` has a CI excluding zero (−7.76 to −0.37) but doesn't survive correction (p_BH≈0.35) — and `self_correct` is an over-predicted, unvalidated move (Figure 9's note), so treat it as suggestive only.

Not shown as a figure: the brand/size_tier mixed-effects results (`mixed_effects_results.csv`) have p-values as small as 1e-127. These are **not a valid population-level significance claim** — brand only truly varies at the model level (3 vs. 3 models), and the model's only random effect is on `instance_id`, which doesn't correct for that. Trust the direction/magnitude of those effects (consistent with everything else in this report), not the p-value.

Figure 13 — Which reasoning steps match the human's, and which don't?

Figure 13. Human-vs-model step alignment: highest-coverage model (left) vs. lowest-coverage model (right), per example

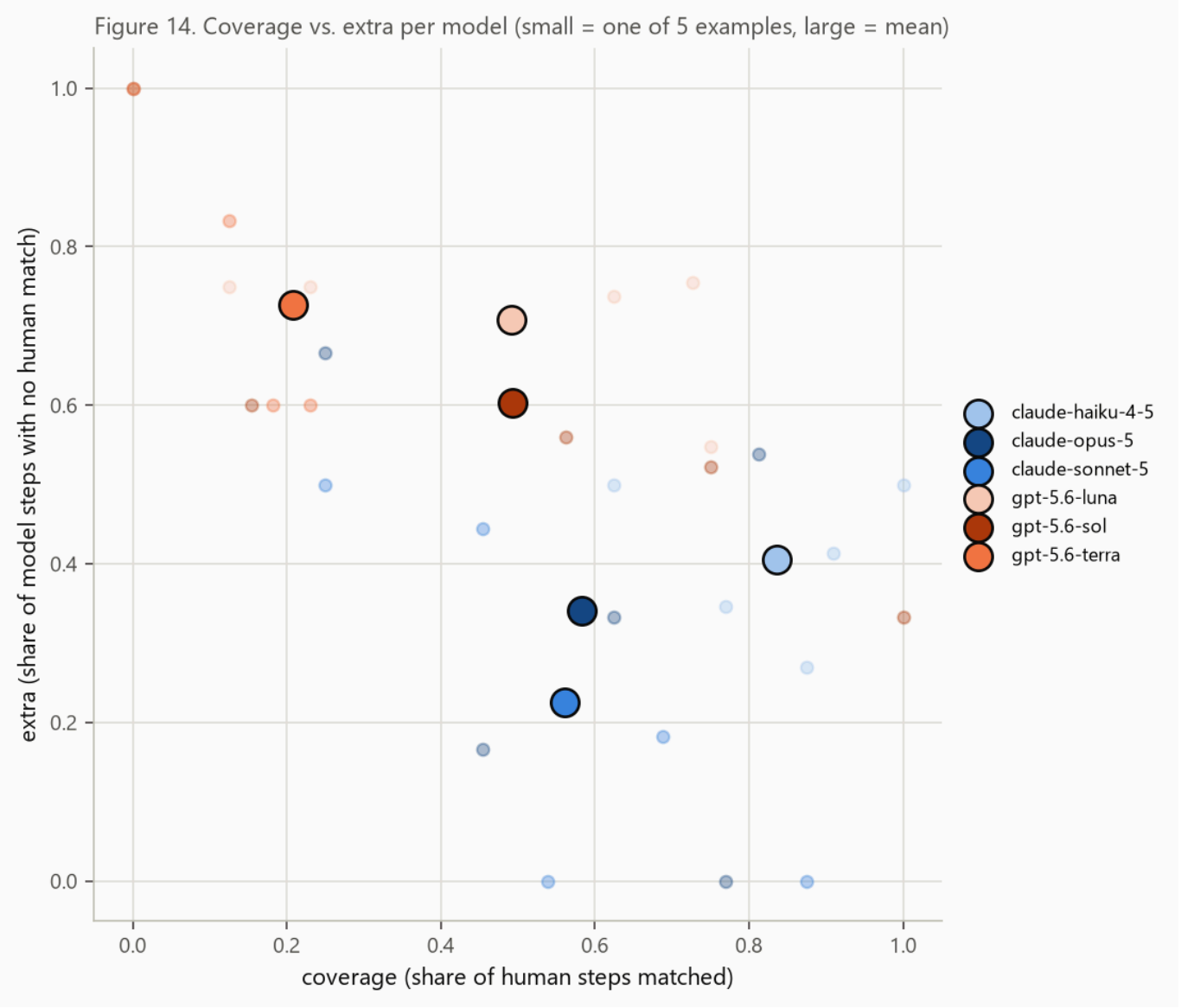




One row per human example, comparing the highest-coverage model (`claude-haiku-4-5`, left) against the lowest-coverage model (`gpt-5.6-terra`, right) — sentence-by-sentence cosine similarity, human sentences on the y-axis, model sentences on the x-axis. Darker cells = closer semantic match. `terra`'s panels are visibly more uniform/lighter — fewer of its sentences closely match any specific human sentence, consistent with its low coverage score below.

Methodological note: the match threshold here (0.703) is **not** the same as Figure 11's revisit threshold (0.884) — reusing the within-trace threshold was tried first and produced 0% coverage everywhere, because cross-author paraphrase similarity (human vs. model, different vocabulary for the same idea) sits structurally lower than within-trace near-duplicate similarity. 0.703 was tuned separately (90th percentile of the pooled human-vs-model similarity distribution) and, like the revisit threshold, should be read as a calibrated choice from a small sample, not a validated ground truth.

Figure 14 — Which models cover the human's points vs. add unrelated content?



Small dots = one of the 5 examples; large outlined dots = per-model mean. `claude-haiku-4-5` **has the highest average coverage (84%)** — the share of human sentences it has *some* matching content for — though this is likely partly a length effect (it also has by far the longest traces, so simply more material to potentially match against). `gpt-5.6-terra` **is the least human-aligned model**: lowest coverage (21%) and highest "extra" (73% of its own sentences match nothing the human said). Separately, `claude-haiku-4-5` also ranks #1 by raw trace-embedding similarity to the human explanation on **all 5 of 5 examples** (0.92–0.94 similarity vs. 0.77–0.91 for every other model) — an independent method reaching the same conclusion as the clustering in Figures 3–4, this time against real external human text rather than internal model-to-model comparison.

Human rationales here are after-the-fact explanations of the gold query (used as the "final answer" analog for the human's own metrics, since a human trace has no `predicted_0.sql` of its own) — not a think-aloud toward an answer still being constructed. Keep that in mind when comparing "human reasoning style" to a model mid-generation. All human-comparison numbers are descriptive only (n=5, no significance tests, per the original analysis spec).

Limitations, in one place

1. **Brand ≈ format**, not cleanly separable in this dataset. Every "brand" claim should be read with that in mind.
2. **Size tier is a price-based proxy** (ranked within brand from `model_pricing.csv`), not a real parameter count.
3. **Brand/size mixed-effects p-values are anti-conservative** (n=3 models per brand) — use effect direction/magnitude, not the raw p-value.
4. **Move-classifier agreement with your labels is moderate**: κ=0.69 (0.76 excluding the two rare moves); the earlier 0.94 was measured against the pre-fix classifier's stored guesses. `restate`/`self_correct` are over-predicted and unvalidated; human sentences were never validated.
5. **Length is a running confound** across move composition, revisit rate, and human-alignment coverage — only partially controlled for (via `log(completion_tokens)` in the mixed-effects models).
6. **n=5 for the human comparison** — descriptive only, no significance tests.
7. **`reasoning_tokens` is OpenAI-only** — Anthropic's API reports no separate reasoning-token count, so it wasn't usable as a cross-brand covariate.
8. **The 47 problems traced by all six models are a harder-than-average slice** (mean accuracy 16.3% on them vs. 29.2% on the other 53; more nested/multi-CTE problems). Figures 3–5 and the human comparison (13–14) use only this subset.
9. **Apostrophe normalization (fixed 2026-09-26)**: OpenAI's text uses curly apostrophes, Anthropic's straight ones. The normalizer now unifies them and every stage was re-run. Effect: Fig 7's OpenAI `i` z-score −39.4 → −32.7 (spurious `ll`/`m` tokens gone); OpenAI's `commitment_action` in Fig 6 ~1.5 → ~1.85. No headline conclusion changed. Previous figures/tables/reports are in `outputs/archive_20260925/`.

For the math, sample sizes, assumptions, axes and justification behind every figure (including a full UMAP explanation and a decoding of the lexical features and reasoning moves), see `figure_guide.md`.

All figures are also in `outputs/figures/` as standalone `.png`/`.svg`; all underlying numbers are in `outputs/tables/`.